Matt Sampson

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Location: Princeton, NJ, USA
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About me

I am a PhD candidate at Princeton university working on developing machine learning tools for the field of computational astrophysics. I am interested in the advancement of machine learning through probability and statistics, and the advancement of science through machine learning.

Technical Skills

Languages	:	Python (JAX,PyTorch, Flax, equinox), R, C++, MATLAB
Mathematical	:	advanced calculus and linear algebra, ODEs/PDEs, statistics and probability, Bayesian inference, computational statistics
Other	:	Git, remote/cluster computing, high-performance computing

Experience

Graduate Researcher (Machine Learning) — 2022 – Present
Princeton University — *Princeton, NJ, USA*

- Designed, implemented, tested score-based diffusion model to assist in galaxy source separation. To be used in upcoming large telescope surveys **Python (JAX, equinox), HPC/GPU computing, remote/cluster computing**
- Implemented a method for computing efficient Hessian free 2nd order methods to quantify hallucination from with neural networks **Python (JAX)**
- Developed automated scripts to extract data from large telescope surveys via intermediate remote clusters **bash,Python**

Honours Researcher — Jan 2021 – Nov 2021
Australian National University — *Canberra, ACT, Australia*

- Assisted in the development of a novel cosmic ray propagation code Criptic (**C++, Python, bash)**
- Troubleshot,fixed bugs and implemented features such as continuous time integration, external file reading, into the source code (**GDB: The GNU Project Debugger**)

Research Assistant (Computational Statistician) — 2019 – 2021
Queensland University of Technology — *Remote – Brisbane, QLD, Australia*

- Assisted in development of statistical tools improving on standard MCMC routines, increase runtime performance by **90%** with no drop model quality **MATLAB, bash**
- Designed and developed statistical analysis tools for very large datasets in biology **Python, R, MATLAB, bash**
- Worked with local cluster computers to perform large scale model fitting of multivariate models

Education

Princeton University — Princeton, NJ, USA
Ph.D. in Computational Astrophysics (machine learning) — *Aug 2022 – Expected mid 2026*

Australian National University — Canberra, ACT, Australia
Honours (similar to US masters) in Computational Astrophysics — *Jan 2021 – Dec 2021*
First class honours, GPA 6.9/7

Queensland University of Technology — Brisbane, QLD, Australia
Bachelor of Mathematics (Applied and Computational) — *Feb 2017 – Nov 2020*
Awarded with distinction GPA 7/7

Queensland University of Technology — Brisbane, QLD, Australia
Bachelor of Science (Physics) — *Feb 2017 – Nov 2020*
Awarded with distinction GPA 6.9/7

Publications

In preparation:

1. **Sampson, M. L.**, Melchior, P. (2023) "Score-matching diffusion models for the generation of the galaxy morphology prior distribution" *Astronomy and Computing*

First author:

1. **Sampson, M. L.**, Beattie, J. R., Krumholz, M. R., Crocker, R. M., Federrath, C., Seta, A. (2022) "The turbulent diffusion of streaming cosmic rays through compressible, partially ionised plasma." *Monthly Notices of the Royal Astronomical Society 519 (1), 1503-1525* **Citations: 18**
2. **Sampson, M. L.**, Melchior, P. (2023) "Spotting Hallucinations in Inverse Problems with Data-Driven Priors" *ICML ML4 Astro*

Second author / significant contributor:

1. Krumholz, M. R., Crocker, R. M., **Sampson, M. L.,** (2022) "Cosmic Ray Interstellar Propagation Tool using Itô Calculus (CRIPTIC): software for simultaneous calculation of cosmic ray transport and observational signatures." *Monthly Notices of the Royal Astronomical Society 517 (1), 1355-1380* **Citations: 6**
2. Stevenson, S., **Sampson, M. L.**, Powell, J., et al. (2019). The impact of pair-instability mass loss on the binary black hole mass distribution. *The Astrophysical Journal, 882(2), 121*. **Citations: 120**

Co-author

1. Beattie, J. R., Krumholz, M. R., Federrath, C., **Sampson, M. L.,** Crocker, R. M. (2022). "Ion Alfvén velocity fluctuations and implications for the diffusion of streaming cosmic rays" *Frontiers in Astronomy and Space Sciences* **Citations: 9**

Metrics all from NASA/ADS: Total of **153 citations** - *h*-index: **4**

Conference Presentations

1. **Sampson, M.** "Spotting hallucinations in inverse problems with data-driven priors". *ICML workshop, Hawaii*, 2023.
2. **Sampson, M.** "Score-based diffusion models for galaxy seperation". *Cosmic Connections, Flatiron Institute, NY*, 2023.
3. **Sampson, M.** "Turbulent diffusion of streaming cosmic rays". *ACAMAR Meeting on Astroparticle Physics, Perth*, 2022.
4. **Sampson, M.** *et. al,*. "Simulating large scale cosmic ray propagation through compressible magnetohydrodynamic turbulence". *Australian Institute of Physics Summer Conference, Brisbane*, 2021.
5. **Sampson, M.** *et. al,*. "Rapid and Effective Population Calibration with modified MCMC". *Australian National University Student Conference, Canberra*, 2020 (best science talk prize).
6. **Sampson, M.** *et. al,*. "The impact of pair-instability mass loss on the binary black hole mass distribution". *Australian National Institute for Theoretical Astrophysics Conference, Melbourne*, 2019.

Teaching

Queensland University of Technology 2018–2020

Experience teaching the following units at QUT where my role was the delivery of content, marking of assignments in an equitable manner, and engaging students with new content such as programming languages and higher level statistics.

SEB104 Grand Challenges in Science (3 Semesters)

SEB113 Quantatative Analysis (3 Semesters)

SEB115 Experimental Science (2 Semesters)

PVB101 Physics of the Very Large (2 Semesters)

BVB204 Ecology - Statistics Component (2 Semesters)

Honors and Awards

- 2022 Princeton Gradute Program First Year Fellowship in Natural Sciences
- 2021 Bok Honours Scholarship in Astrophysics at the RSAA
- 2020/2019/2018 Deans List of Commendation for academic excellence QUT
- 2019 ICRAR-Pawsey Centre Vacation Scholar
- 2018 Swinburne Centre for Astrophysics and Supercomputing Summer Vacation Scholarship

References

Provided as per request